

Yurou Celine Xiao

 LinkedIn

 Google Scholar

 Ionic Stacks



SKILLS

Process & Hardware Design

- High-pressure high-temperature system design, pilot scaling, SolidWorks, engineering drawings, GD&T, laser cutter, drill press, soldering, electronics assembly

Electrochemistry & Fabrication

- LANDT Battery Test Systems, Metrohm Autolab, CC/CV, LSV, EIS, catalyst synthesis, electrode fabrication, electrolyte formulation

Analytical Chemistry & Software

- SEM, XRD, GC, IC, NMR, UV-vis, Python, MATLAB, OriginPro

EDUCATION

Ph.D. Mechanical Engineering – University of Toronto

- Electrochemical systems for integrated capture and conversion of CO₂

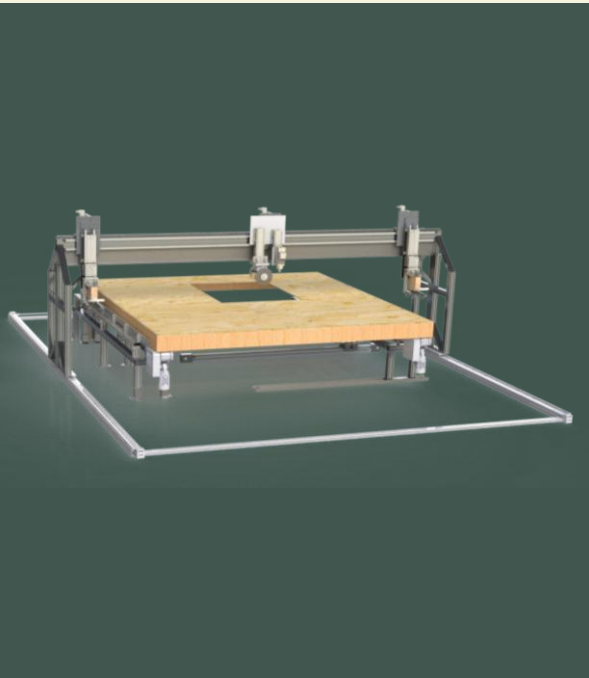
B.A.Sc. Mechanical Engineering – University of Alberta

Industry

MAY 2017 – AUG 2017

MODULAR CONSTRUCTION

Designed and modelled an industrial multi-function CNC machine to automate fastening and cutting processes.



SEPT 2017 – DEC 2017

SYNCRUDE CANADA LTD.

Engineered emergency and scheduled maintenance reliability packages for O&G and utility plants.



MAY 2018 – DEC 2018

CITY OF EDMONTON

Acted as lead delegate in site meetings for municipal infrastructure projects.



AUG 2019 – DEC 2019

WSP CANADA

Transformed legacy P&IDs into a 3D-linked compliance tool and authored core engineering documentation.



Start-up

SEP 2020 – DEC 2020

CERT SYSTEMS INC. 

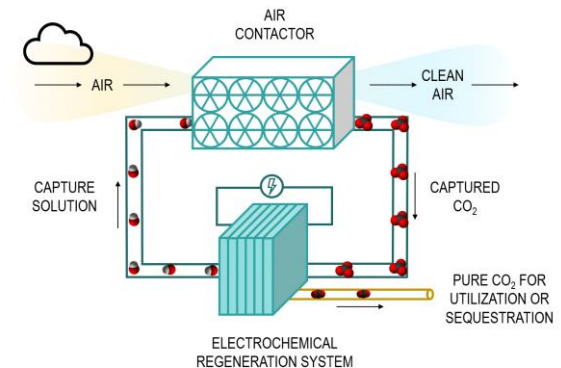
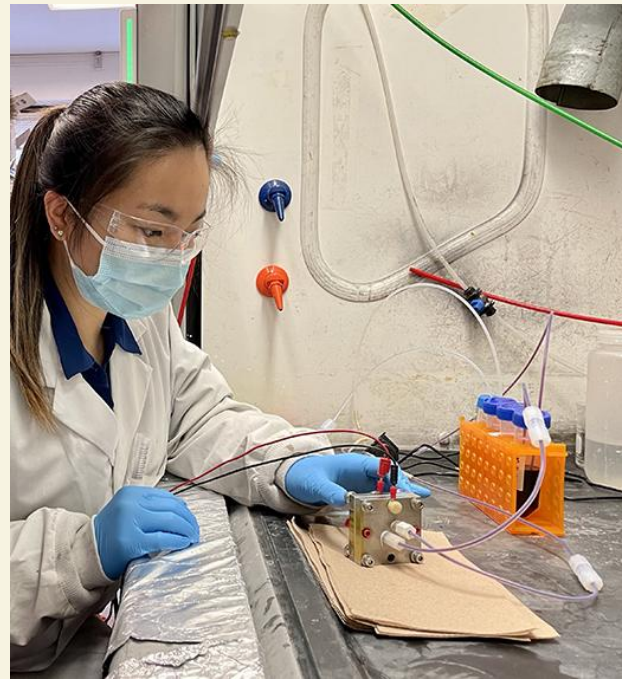
Scaled the electrochemical conversion of CO₂ into ethylene by operating a pilot plant (up to 40,000 cm² total active area), contributing to Team CERT's success as a Top 10 Finalist in the NRG COSIA Carbon XPRIZE.



SEP 2021 – OCT 2022

E-QUESTER 

Co-founded a startup team that developed a low-cost and scalable electrically powered direct air capture (DAC) system for CO₂ removal, resulting in a \$250,000 Student Prize and named a Top 60 Milestone Award Finalist in the XPRIZE Carbon Removal competition.



Research

MAY 2020 – MAR 2021

**UNIVERSITY OF
CALGARY**

Optimized energy performance for the net-zero MacKimmie Tower through thermal data analysis.



JAN 2024 – APR 2024

**CSIRO ENERGY
CENTER**

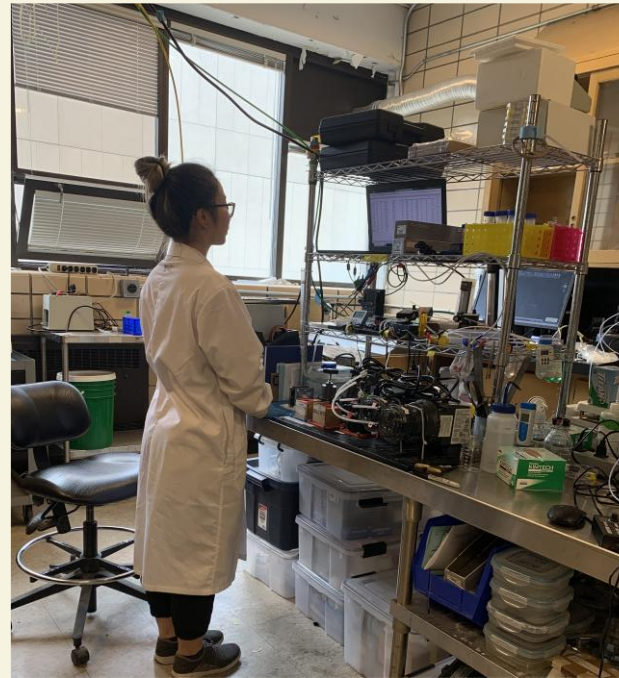
Developed and scaled CO₂ electrolyzers from 1 cm² to 100 cm² for integrated CO₂ capture and conversion systems.



MAY 2023 – AUG 2025

**SINTONLAB:
PROGRAM LEAD**

Directed a multi-year, \$2M+ industry sponsored research program on integrated capture and conversion of CO₂.



SEP 2023 – AUG 2025

**SINTONLAB:
ELECTROLYZER DESIGNER**

Managed electrolyzer Design for Manufacturing (DFM) as the technical liaison between R&D and machining teams.

